

Module 03: Perception in Embodied

Learning Objectives

1. Define **Perception** within the context of Embodied.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In Embodied, we see Perception manifest in: * **Specific Example 1:** A chess grandmaster glances at a board mid-game and the "right" move presents itself instantly as a perceptual gestalt -- she does not calculate sequences but sees the board configuration as a coherent pattern, and the slight buzz of recognition in her body confirms the match between the current visual array and the deep priors accumulated from tens of thousands of prior games stored in her generative model. * **Specific Example 2:** An experienced martial artist perceives a subtle shift in his opponent's shoulder and instinctively reads it as the precursor to a specific strike; this intuitive perception operates at the Markov blanket with millisecond precision, where the generative model's prediction of the opponent's next move is so refined by training that the incoming visual signal is immediately completed by the model into a full action trajectory, experienced not as inference but as direct bodily seeing.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.